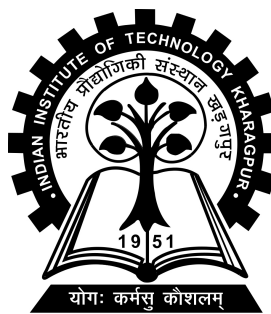


Singularity Free Combined Vector Field Based Obstacle Avoidance and Path Following

Project-II (AE47006) report submitted to
Indian Institute of Technology Kharagpur
in partial fulfilment for the award of the degree of
Dual Degree (B.Tech + M.Tech)
in
Aerospace Engineering

by
Dheeraj
(22AE30009)

Under the supervision of
Professor Sikha Hota



Department of Aerospace Engineering
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Spring Semester, 2025-26

May 4, 2026

DECLARATION

I certify that

- (a) The work contained in this report has been done by me under the guidance of my supervisor.
- (b) The work has not been submitted to any other Institute for any degree or diploma.
- (c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.
- (d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

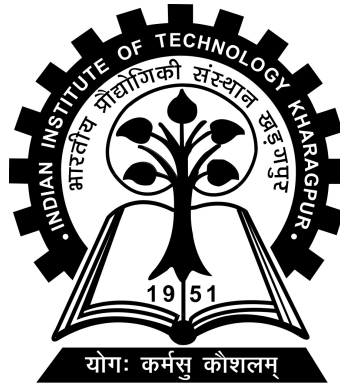
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CERTIFICATE

This is to certify that the project report entitled “Singularity Free Combined Vector Field Based Obstacle Avoidance and Path Following” submitted by Dheeraj (Roll No. 22AE30009) to Indian Institute of Technology Kharagpur towards partial fulfillment of requirements for the award of the degree of Dual Degree (B.Tech + M.Tech) in Aerospace Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

Date: May 4, 2026
Place: Kharagpur

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Abstract

Name of the student: **Dheeraj**

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Degree for which submitted: **Dual Degree (B.Tech + M.Tech)**

Department: **Department of Aerospace Engineering**

Thesis title: **Singularity Free Combined Vector Field Based Obstacle Avoidance and Path Following**

Thesis supervisor: **Professor Sikha Hota**

Month and year of thesis submission: **May 4, 2026**

A Combined Vector Field (CVF)-based framework for unmanned aerial vehicle (UAV) path following and obstacle avoidance in a two-dimensional environment is presented in this thesis. A unified and geometry-driven formulation of the guidance issue is made possible by the use of twice differentiable implicit functions to describe the obstacle bounds and the desired path. Conservative (gradient-based) and solenoidal (orthogonal) components are combined using nonlinear, state-dependent weighting functions to create a CVF-based path-following vector field that guarantees convergence to the intended path while preserving smooth advancement along it. Repulsive and circulatory behaviors around obstacle borders are produced by extending a similar approach to obstacle avoidance.

A structured blending technique based on guidance direction (course angle) combination is suggested to solve the problem of singularities resulting from direct vector field superposition. To enable seamless transitions between obstacle-avoidance and path-following behaviors, a distance-dependent weighting mechanism is provided.

MATLAB simulations are used to implement and assess the suggested framework in a variety of scenarios, such as circular and random pathways in the presence of various barrier configurations and wind disturbances. The outcomes show that the technique may accomplish efficient path following while avoiding obstacles in a variety of scenarios.

Limitations pertaining to parameter sensitivity, edge-case behaviors, and the field combination strategy's robustness are also noted in the paper. These findings point to future research directions, such as enhanced blending mechanisms, expansion to higher-dimensional systems, and experimental confirmations.

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Abbreviations

UAV	U n m anned A erial V ehical
MAV	M iniature A ir V ehicle
VF	V ector F ield
PF	P otential F ield
VFF	V irtual F orce F ield
PFVF	P ath F ollowing V ector F ield
LVF	L yapunov V ector F ield
GVF	G radient V ector F ield
CVF	C ombined V ector F ield
PI	P roportional I ntegral
PD	P roportional D erivative
PID	P roportional I ntegral D erivative

Symbols

g	Gravitational Acceleration ($= 9.81 \text{ m/s}^2$)
ϕ_{max}	Maximum allowed Bank Angle
ψ	Heading Angle
ψ_d	Desired Heading Angle
χ	Course Angle
χ_d	Desired Course Angle
χ_{path}	Course Angle Determined solely from the Path following VF
χ_{obs}	Course Angle Determined solely from the Obstacle VF
δ	Safety Margin from the Obstacle
α	Obstacle and Path VF Switching factor

Chapter 1

Introduction

Unmanned Aerial Vehicles (UAVs) have emerged as the critical technology across a wide range of applications, including surveillance, environmental monitoring, infrastructure inspection, and target tracking [1–4]. Many of these applications fundamentally depend on a UAV’s ability to autonomously follow a prescribed path while maintaining safe operation in environments that may contain obstacles and no-fly zones. This requirement introduces a dual objective: accurate path following and reliable obstacle avoidance, both of which must be achieved under vehicle dynamic constraints and limited on-board computational resources.

Traditionally, UAV guidance systems rely on waypoint-based navigation, where a trajectory is discretized into a sequence of points generated offline and transmitted to the vehicle [5–14]. These waypoints are tracked using control laws designed for the specific UAV platform [15–17]. While this approach is widely adopted in commercial autopilot systems such as Pixhawk and Piccolo [18–20], it suffers from several inherent limitations. In particular, waypoint-based methods are not well suited for environments where obstacles are detected after mission planning, as they require partial or complete trajectory replanning [21, 22]. Moreover, the use of discrete waypoints can lead to large cross-track errors, missed waypoints under disturbances such as wind, and trajectories that do not fully respect vehicle kinematic constraints.

To address these challenges, Vector Field-based (VF) guidance methods have been proposed as an alternative framework for UAV navigation. Unlike waypoint-based approaches, VF methods provide continuous guidance at every point in space, enabling the UAV to respond in real time to environmental challenges without requiring explicit replanning. This makes them particularly suitable for autonomous operation in uncertain and dynamic environments.

Early VF-based obstacle avoidance methods were developed using artificial Potential Fields (PF) and Virtual Force Fields (VFF) [3, 23–32]. These methods construct artificial attractive forces toward a goal and repulsive forces away from an obstacle, resulting in a velocity vector, derived from the gradient of a scalar potential function. While computationally efficient and intuitive, PF and VFF methods suffer from several limitations. These include the presence of local minima, where the vehicle can become trapped, and oscillatory behavior in confined environments such as narrow corridors [33–35]. Various strategies have been proposed to mitigate these issues, including virtual waypoints [36], escape route methods [37], and navigation functions [22]. However, these methods remain fundamentally goal-oriented and therefore do not inherently ensure convergence back to the desired path after obstacle avoidance. As a result, PF- and VFF-based approaches can produce trajectories that deviate significantly from the intended path.

To overcome these shortcomings, research has shifted toward Path Following Vector Fields (PFVFs), which explicitly encode both convergence to the path and motion along it. Two major classes of PFVFs have been extensively studied: Lyapunov Vector Fields (LVF) [3, 26–29, 38–43] and Gradient Vector Fields (GVF) [30–32, 44]. LVFs are constructed from Lyapunov functions that capture the geometry of the path, enabling provable convergence and stability for specific path shapes, such as straight lines and circles. Extension of LVF methods to more complex trajectories has been proposed [29], although they often require switching between multiple Lyapunov functions, which can increase system complexity.

In contrast, GVFs provide a more general and flexible framework by defining paths as the intersection of the implicit surfaces. The resulting vector field is composed of two decoupled components: a convergence term, derived from the gradient of a potential function, and a circulation term, which generates motion along the path through cross products of surface gradients. This decomposition allows independent tuning of convergence and tangential behavior, and naturally extends to higher-dimensional and time-varying paths [30–32]. Under appropriate conditions, GVFs on the implicit functions guarantee convergence to the desired path, making them a powerful tool for UAV guidance.

Despite these advantages, integrating obstacle avoidance into PFVF frameworks remains a nontrivial problem. A common approach is to construct a repulsive vector field representing the obstacle and combine it with the path following field [45]. However, superposing vector fields introduces several challenges, including sensitivity to weight selection and the presence of singularities where opposing vectors cancel, leading to null guidance. Such singularities can lead to undesirable behaviors, including trap conditions in which the UAV becomes stationary or fails to avoid an obstacle [46]. Additionally, improper tuning of field parameters may result in excessive deviation from the original path or inefficient avoidance trajectories.

Accordingly, this thesis proposes a Combined Vector Field (CVF) framework for UAV guidance that builds on the principles of Path Following Vector Fields (PFVFs) and Gradient Vector Fields (GVFs), introducing a structured, nonlinear formulation for improved robustness. The proposed approach constructs the guidance field by decomposing it into a conservative component that ensures convergence to the desired path and a circulatory (orthogonal) component that induces motion along the path. Unlike classical GVF formulations that rely on linear superposition, the proposed method employs a nonlinear state-dependent weighting function to enable smooth transitions between convergence and circulation behaviors, thereby reducing sensitivity to parameter tuning [47, 48]. This formulation is further extended to obstacle avoidance by defining a similar CVF around obstacle boundaries, generating both

repulsive and circulatory effects. To mitigate the singularities, the path-following and obstacle-avoidance vector fields are combined through structural blending of guidance directions. Based on this formulation, a guidance algorithm is proposed to achieve singularity-free path-following with obstacle avoidance.

Chapter 2

Singularity Free Combined Vector Field Based Path Following and Obstacle Avoidance

This chapter presents the formulation of the Combined Vector Field (CVF) approach for path following and obstacle avoidance in 2D. A singularity-free vector field is then constructed based on this formulation. Subsequently, a guidance algorithm for path following and obstacle avoidance is proposed. Finally, MATLAB simulations are presented to evaluate the performance of the proposed approach across different paths and obstacle configurations.

2.1 Problem Description

In the 2D path following and obstacle avoidance, the UAV is considered to have altitude hold with zero climb rate and a velocity hold for constant airspeed (V_a). The kinematic model of the UAV in the inertial XY-frame, as in the figure [2.1](#), can

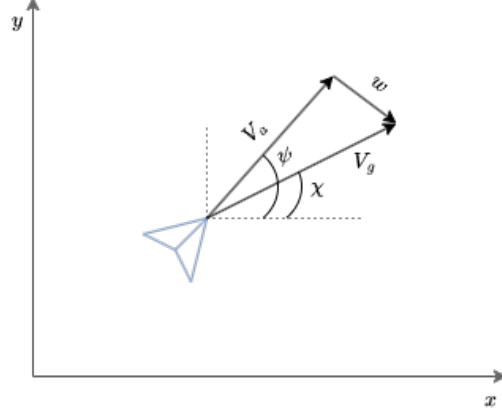


FIGURE 2.1: 2D Kinematics of the UAV

be given as,

$$\dot{x} = V_a \cos(\psi) + w_x = V_g \cos(\chi) \quad (2.1)$$

$$\dot{y} = V_a \sin(\psi) + w_y = V_g \sin(\chi) \quad (2.2)$$

$$\dot{\chi} = u_1 \quad (2.3)$$

where,

$\mathbf{p} = [x, y]^T$ is the position of the UAV in the 2D inertial frame

$\mathbf{v} = [\dot{x}, \dot{y}]^T$ are the inertial velocities.

ψ is the UAV heading angle.

χ is the course angle.

V_a is the constant airspeed.

V_g is the ground speed.

$w = [w_x, w_y]^T$ are the inertial wind velocities.

u_1 is the commanded course rate satisfying the constraint $|u_1| \leq \dot{\chi}_{max}$

Additionally, it can be observed that the relationship between airspeed and ground speed can be expressed as follows,

$$V_g = (w_x \cos \chi + w_y \sin \chi) + \sqrt{V_a^2 - (w_x \sin \chi - w_y \cos \chi)^2} \quad (2.4)$$

We have a desired 2D arbitrary twice differentiable curved path, denoted as $C_1 : f(x, y) = 0$, that the UAV should follow, and an obstacle to avoid, denoted as $C_2 : g(x, y) = 0$ with a safety margin of (δ) . We define the directional distances from the current UAV position $\mathbf{p}$ to these paths as $r = f(x, y)$ and $r_{obs} = g(x, y)$, respectively. A large value of $|r|$ indicates that the UAV is far from the desired path, while a large value of r_{obs} suggests that the UAV is far from the obstacle. When $r = 0$, it means the UAV is exactly on the desired path, whereas $r_{obs} = 0$ indicates that the UAV has collided with the obstacle.

The 2D arbitrary twice differentiable curved path following and obstacle avoidance problem can be summarized as follows: To design the input command u_1 for the UAV whose kinematics are given by (2.1)-(2.3), such that,

- (i) The input constraint $|u_1| \leq \dot{\chi}_{max}$ holds.
- (ii) and if possible, force the UAV to fly along the curve C_1 , and if not, force it to make the absolute distance $|r|$ as small as possible, while avoiding the obstacle C_2 with a safety margin of (δ) .

2.2 Combined Vector Field

The Helmholtz theorem states that any arbitrary vector field can be decomposed into a conservative and a solenoidal part. The radial part is irrotational (Curl-free) and forces the UAV towards or away from the desired path/obstacle. The solenoidal part is purely rotational, forcing the UAV to align with the curve's direction. The

conservative and solenoidal parts of the vector field can be determined by obtaining the gradient of the curve equation ∇f and taking the vector orthogonal of the conservative field, respectively.

The conservative part after the normalization becomes,

$$\mathbf{v}_c = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_x \\ f_y \end{pmatrix} \quad (2.5)$$

and the solenoidal part after normalization becomes,

$$\mathbf{v}_s = \frac{1}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} f_y \\ -f_x \end{pmatrix} \quad (2.6)$$

Note: It can also be seen from the (2.5) and (2.6) that both conservative and solenoidal fields are normal to each other at every point in space.

Path following Combined Vector Field Now, according to Helmholtz's decomposition theorem, a certain combination of two perpendicular vector fields gives a new vector field. Furthermore, if we construct the new vector field from the conservative and solenoidal components such that,

- (i) When $\mathbf{p}$ is located very far away, i.e., $|r|$ is huge, then the new vector field aligns with the conservative field and forces the UAV to move directly towards the curve C_1 .
- (ii) When $\mathbf{p}$ is on the curve C_2 , i.e., $r = 0$, then the new vector field is totally aligned with the solenoidal vector field, which forces the UAV to follow and remain on the curve C_2 .

Hence, the new combined vector field for path following can be given as,

$$\mathbf{v}_{path} = \begin{pmatrix} \dot{x}_{path} \\ \dot{y}_{path} \end{pmatrix} = -V_g \tanh(\kappa r)(\mathbf{v}_c)_{path} + sV_g \text{sech}(\kappa r)(\mathbf{v}_s)_{path} \quad (2.7)$$

$$= \frac{V_g}{\sqrt{f_x^2 + f_y^2}} \begin{pmatrix} -f_x \tanh(\kappa f(x, y)) + s f_y \text{sech}(\kappa f(x, y)) \\ -f_y \tanh(\kappa f(x, y)) - s f_x \text{sech}(\kappa f(x, y)) \end{pmatrix} \quad (2.8)$$

where $\kappa > 0$ is a path following vector field design parameter. The larger κ , the faster the convergence, i.e., the conservative part dominates for the most part. The parameter $s = \pm 1$ determines the solenoidal vector field. The convergence of this derived combined vector field is proved in [48].

Obstacle Combined Vector Field Similarly, for the obstacle combined vector field, the conservative and solenoidal parts of the field can be generated as,

$$(\mathbf{v}_c)_{obs} = \frac{1}{\sqrt{g_x^2 + g_y^2}} \begin{pmatrix} g_x \\ g_y \end{pmatrix} \quad (2.9)$$

$$(\mathbf{v}_s)_{obs} = \frac{1}{\sqrt{g_x^2 + g_y^2}} \begin{pmatrix} g_y \\ -g_x \end{pmatrix} \quad (2.10)$$

Now, we construct the new vector field from the conservative and solenoidal vector field components of the obstacle, such that, when $\mathbf{p}$ is located near the obstacle, i.e., $|r_{obs}|$ is small, then the conservative term dominates such that it pushes the UAV away from the obstacle, and as the distance from the obstacle increases, the conservative term diminishes, and the solenoidal term appears.

The above-described vector field can be generated as follows,

$$\mathbf{v}_{obs} = \begin{pmatrix} \dot{x}_{obs} \\ \dot{y}_{obs} \end{pmatrix} = V_g \text{sech}(\kappa_{obs} r_{obs})(\mathbf{v}_c)_{obs} + s_o V_g \tanh(\kappa_{obs} r_{obs})(\mathbf{v}_s)_{obs} \quad (2.11)$$

$$= \frac{V_g}{\sqrt{g_x^2 + g_y^2}} \begin{pmatrix} g_x \text{sech}(\kappa_{obs} g(x, y)) + s_o g_y \tanh(\kappa_{obs} g(x, y)) \\ g_y \text{sech}(\kappa_{obs} g(x, y)) - s_o g_x \tanh(\kappa_{obs} g(x, y)) \end{pmatrix} \quad (2.12)$$

Where, again, $\kappa_{obs} > 0$ is the obstacle combined vector field design parameter. A smaller κ_{obs} means the vector field is strongly influenced by the conservative term near the obstacle. The parameter $s_o = -s$ determines the direction of the solenoidal field, i.e., the obstacle vector field curls in the opposite sense to the path-following vector field.

The parameter s_o will be extensively used and manipulated to develop a suitable algorithm for path following and obstacle avoidance.

2.3 Singularity-free Combined Vector Field

When two arbitrary vector fields are combined by vector addition, there may be points where the two vectors of the respective fields completely cancel each other, i.e., they are of the same length and opposite in direction. Such points are termed as **null points** or **singularities** in the vector field.

Since the guidance of the UAV is completely dependent on the desired direction at the given point, from the vector field. This implies that at the singularities, the UAV will have no desired direction/guidance.

To avoid the singularities, we can add the two vector fields not by vector addition but by adding the angles from the inertial x-axis of the respective vectors with some weighting factors dependent on the UAV's relative position to the obstacle.

The weighting factor wtf is designed, and the angles are combined in such a way that when the UAV is very far away from the obstacle, the weighting factor becomes *Zero*, i.e., the angle from the path following vector field remains, and say when the UAV moves toward the obstacle the weighting factor smoothly rises and becomes unity, i.e., the UAV reaches to the perpendicular distance of δ from the obstacle [49, 50].

$$wtf = \operatorname{sech} \left(\frac{PN - \delta}{\alpha} \right) \quad (2.13)$$

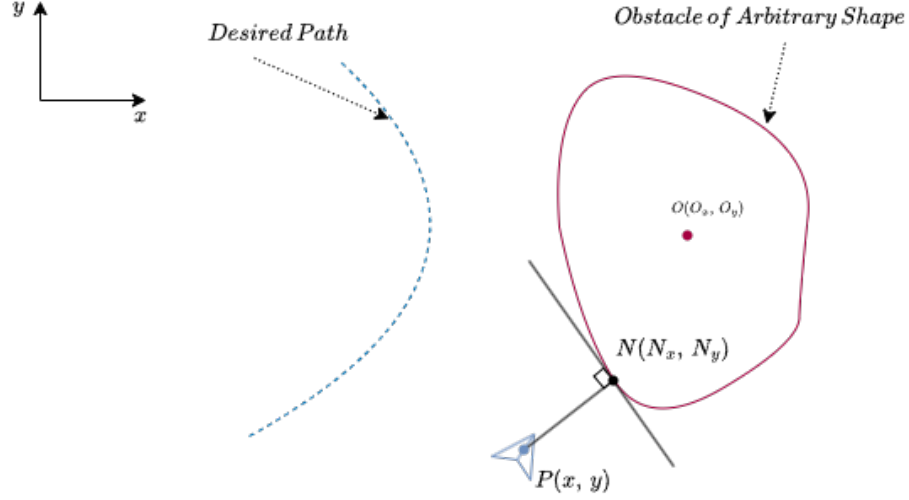


FIGURE 2.2: Smallest Perpendicular from UAV to Obstacle

where PN is the minimum perpendicular distance from the UAV to the obstacle. δ is the safety distance from the obstacle. α is the parameter that dictates the rate of switching between the fields.

Then, the singularity-free combined path following and obstacle avoidance vector field can be given as,

$$\chi_d = \operatorname{wrapTo2Pi}((1 - wtf) \times \chi_{path} + wtf \times \chi_{obs}) \quad (2.14)$$

where χ_{path} and χ_{obs} are the desired course angles given by the path following vector field and obstacle vector fields alone, respectively.

This strategy of angle addition negates the presence of any singularity in the field, while *sech* weighting factor conserves the fields and smoothly merges two fields with each other. However, some quirks related to the wrapping behavior, the sensitivity

of the switching parameter (α), and the formation of loops and premature stable points due to the selection of some pairs of s and s_o still remain.

2.4 Algorithm Based on Singularity-free Combined Vector Field

- Define a twice differentiable 2D implicit desired path to follow $f(x, y) = 0$ and another twice differentiable 2D implicit obstacle $g(x, y) = 0$.
- Compute and normalize the conservative and solenoidal vector fields for both path and obstacle curves using Eqs. (2.5), (2.6), (2.9), and (2.10).
- Compute the Combined Vector Fields for both Path-following and obstacle avoidance separately using Eqs. (2.8), (2.12).
- Compute the angles χ_{path} and χ_{obs} corresponding to the Path following and Obstacle vector fields, respectively. Using

$$\chi_{path} = \text{atan2}((\dot{y}_d)_{path}, (\dot{x}_d)_{path}) \quad (2.15)$$

$$\chi_{obs} = \text{atan2}((\dot{y}_d)_{obs}, (\dot{x}_d)_{obs}) \quad (2.16)$$

$$(2.17)$$

Remember The function $\text{atan2}()$ has the range $(-\pi, \pi]$.

- **CASE I:** The extended obstacle (obstacle with safety margin) is completely outside the desired path, i.e., the obstacle lies completely in the region where pseudo distance $r = f(x, y)$ is strictly positive ($r > 0$).
 - Compute an angle (ζ) of the vector from the foot of the smallest perpendicular from the obstacle center of mass to the desired path (C_1) and the

obstacle center of mass.

$$\zeta = \text{wrapTo2Pi}(\text{atan2}(O_y - M_y, O_x - M_x)) \quad (2.18)$$

where O is the obstacle center of mass, and M is the foot of the smallest perpendicular from the obstacle center of mass to the desired path (C_1). while O_x, O_y and M_x, M_y are the x and y components of the positions at O and M respectively.

- Then, wrap both the χ_{path} and χ_{obs} to $[\zeta, \zeta + 2\pi)$. Which in turn switches the angle wrapping point from $-\pi$ (due to the fact that the initial angle comes from the $\text{atan2}()$) to a chosen point ζ .

$$\chi_{path} = \zeta + \text{wrapTo2Pi}(\chi_{path} - \zeta) \quad (2.19)$$

$$\chi_{obs} = \zeta + \text{wrapTo2Pi}(\chi_{obs} - \zeta) \quad (2.20)$$

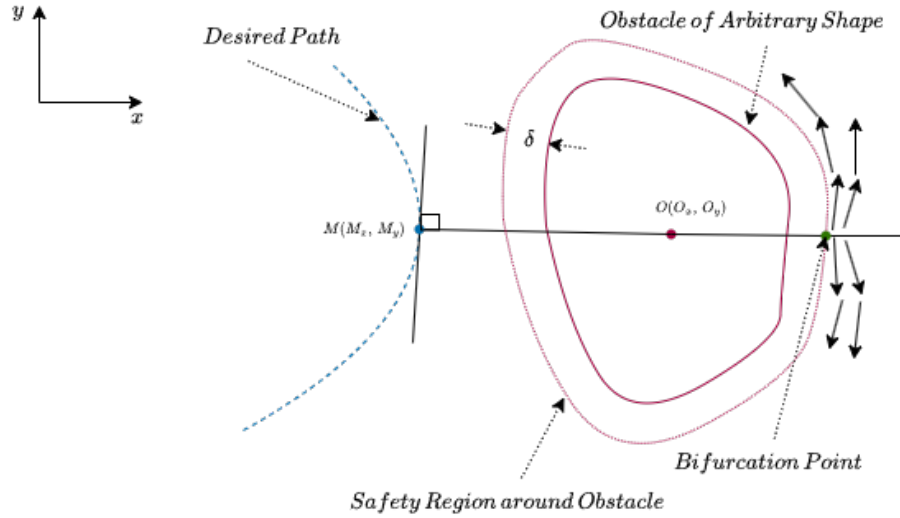


FIGURE 2.3: Smallest Perpendicular on Path from Obstacle COM

NOTE: It is observed from simulations that the vector fields bifurcate (vector points in different directions around that point), around that angle

wrapping point outside the obstacle. To make use of this fact, we move the bifurcation point to ζ such that the UAV travels approximately the same distance to the curve from either side of the obstacle.

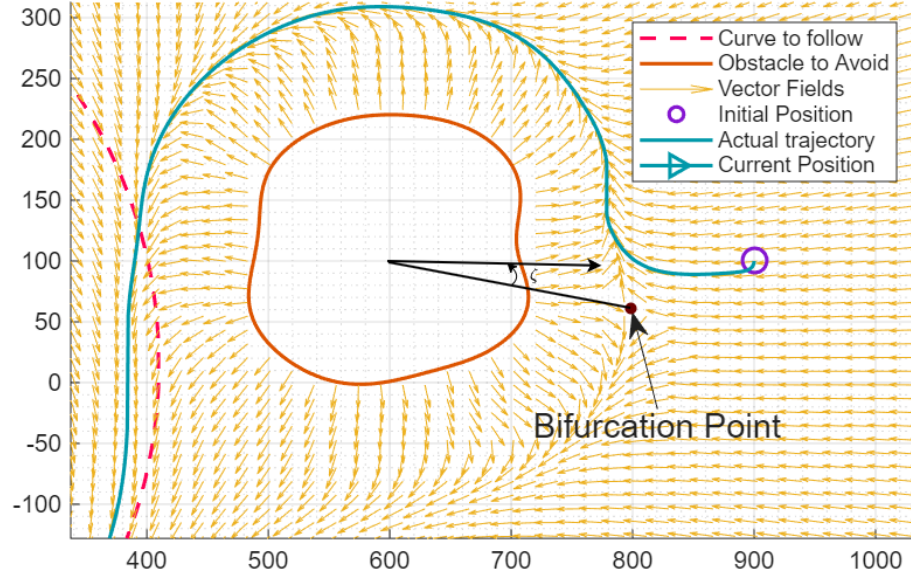


FIGURE 2.4: Angle wrapping point/Bifurcation point in the singularity-free combined vector field

- **CASE II:** The obstacle lies in such a way that the extended obstacle (with the safety margin) intersects the desired path, i.e., making one half of the obstacle inside the Path ($r < 0$) and the other half outside ($r > 0$).
- **Part of the obstacle is outside ($r \geq 0$):** For this part, we want to enforce the obstacle's solenoidal part to curl opposite to the path. For that, the pseudo code can be given as,

Algorithm 1 Pseudo code to handle the path following and obstacle avoidance for the part of the obstacle outside the desired path

- 1: **if** $s_o = 1$ and $\chi_{path} < \chi_{obs}$ **then**
 - 2: Add 2π to the current value of χ_{path}
 - 3: **else if** $s_o = -1$ and $\chi_{obs} < \chi_{path}$ **then**
 - 4: Add 2π to the current value of χ_{obs}
 - 5: **end if**
-

- **Part of the obstacle is inside** ($r < 0$): Similarly, for this part, we want to enforce the obstacle's solenoidal part to curl similar to the path.

For that, the pseudo code can be given as,

Algorithm 2 Pseudo code to handle the path following and obstacle avoidance for the part of the obstacle inside the desired path

```

1: if  $s_o = -1$  and  $\chi_{path} < \chi_{obs}$  then
2:   Add  $2\pi$  to the current value of  $\chi_{path}$ 
3: else if  $s_o = 1$  and  $\chi_{obs} < \chi_{path}$  then
4:   Add  $2\pi$  to the current value of  $\chi_{obs}$ 
5: end if

```

The above algorithms ensure that whenever the UAV's path intersects an obstacle, it crosses the obstacle without becoming stuck at any premature stable points or loops.

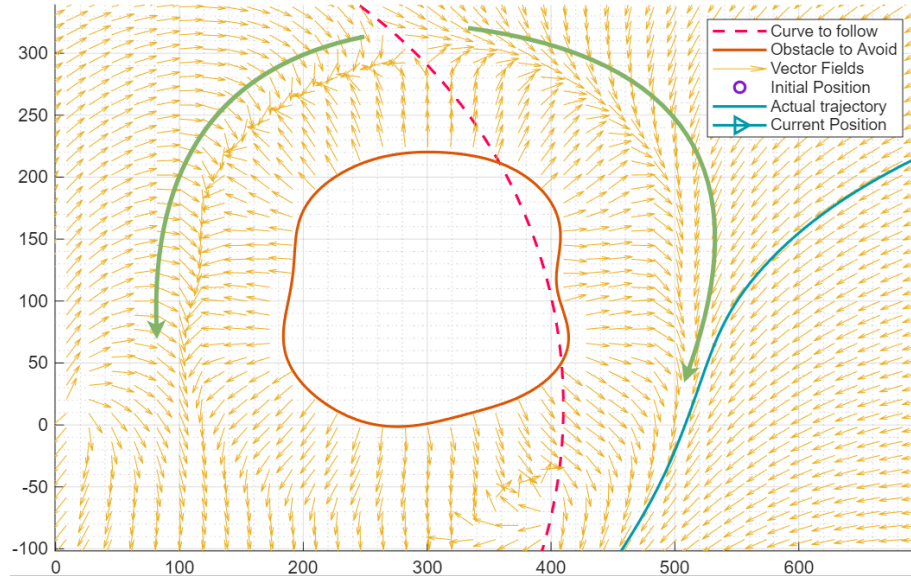


FIGURE 2.5: Vector Field around the obstacle, when the obstacle lies on the desired path

- **CASE III:** The extended obstacle is totally inside the desired path, i.e., $r < 0$ around the whole extended obstacle. Then the **same algorithm** is applied as in **CASE I**.

- Now, compute the Singularity-free Combined Vector Field for Path following and Obstacle avoidance using (2.14).
- Design a controller which moves the UAV to directions obtained from the singularity-free CVF.

The proportional course rate controller for the constrained UAV based on the above-derived singularity-free combined vector field. The desired course determined by the combined vector field in (2.14) can be computed as

$$\chi_d = \arctan2(\dot{y}_d, \dot{x}_d) \quad (2.21)$$

where $\arctan2(\cdot, \cdot) \in (-\pi, \pi]$ and course error $\chi_e = \langle \chi - \chi_d \rangle$, where $\langle \cdot \rangle$ is the $(-\pi, \pi]$ angle wrapper.

We can now design a proportional course rate controller for the path following and obstacle avoidance as,

$$u_1 = \omega_1, \quad \text{with} \quad |\omega_1| \leq \dot{\chi}_{max} \quad (2.22)$$

where $\omega_1 = -k_\chi \chi_e$ and $k_\chi (> 0)$ is the proportional feedback gain.

2.5 Limitations

This approach to creating a singularity-free combined vector field by using angle addition and ensuring that the solenoidal part of the obstacle vector field has a specific direction of rotation is effective in most cases; however, it has some limitations.

- (i) **Inherent CVF Limitations** There is no common κ and κ_{obs} for different curves. The vector field design parameter should be changed accordingly for different curves.
- (ii) **Limitations of the Algorithm**

- **Edge Cases** In the case when the obstacle is on the desired path itself (CASE II), then some artifacts in the vector field near the obstacle appear due to the strict implication of the rotation of the solenoidal part of the obstacle VF in the specific direction. (which is done by adding 2π to the χ_{path} or χ_{obs}).

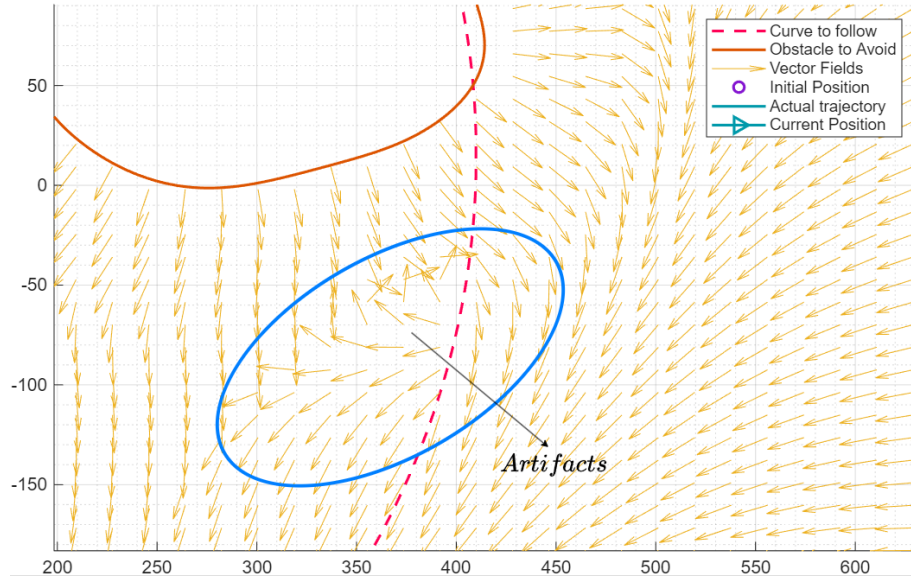


FIGURE 2.6: Artifacts in the vector field appearing near the obstacle

- We cannot reliably bifurcate the vector fields at a wrapping point (ζ). Also, bifurcation may become dangerous, as there may be multiple bifurcation points (depending on the geometry of the obstacle/path), which can create a premature stable point or a loop.
- The vector field is sensitive to parameter (α), which governs how rapidly the effect of the obstacle VF diminishes or remains. If the parameter α is large, then the obstacle VF will have an effect on points that are very far away. which may again lead to the UAV's inability to converge, i.e., premature stable points and/or loops.

Note: The following limitations are identified based on simulation studies.

2.6 Simulations

2.6.1 Circular Desired Path and a Circular Obstacle

For the 2D path following of a twice differentiable circular path with a twice differentiable circular obstacle with singularity-free CVF-based guidance, the desired path is given by

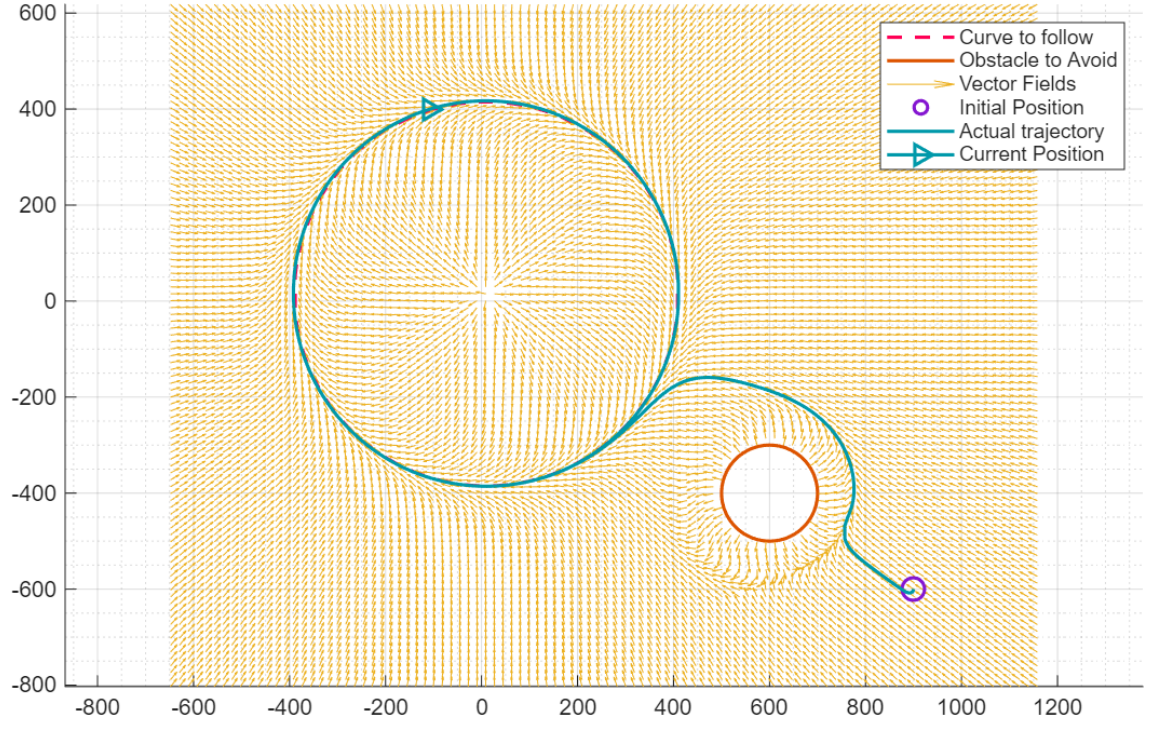
$$C_1 : \quad f(x, y) = (x - 10)^2 + (y - 15)^2 - 400^2 \quad (2.23)$$

with constant airspeed of 15 m/s , initial course angle $-\pi/2$, rotational sense of the solenoidal vector field for path-following ($s =$) is 1 (clock-wise sense), Path-following vector field design parameter ($\kappa =$) is 20, Obstacle vector field design parameter ($\kappa_{obs} =$) is 5, proportional feedback gain ($k_\chi =$) is 1.5 and the maximum allowable roll angle is 45° which limits the course rate. Additionally, simulation is done in the presence of wind $[6, 8] \text{ m/s}$.

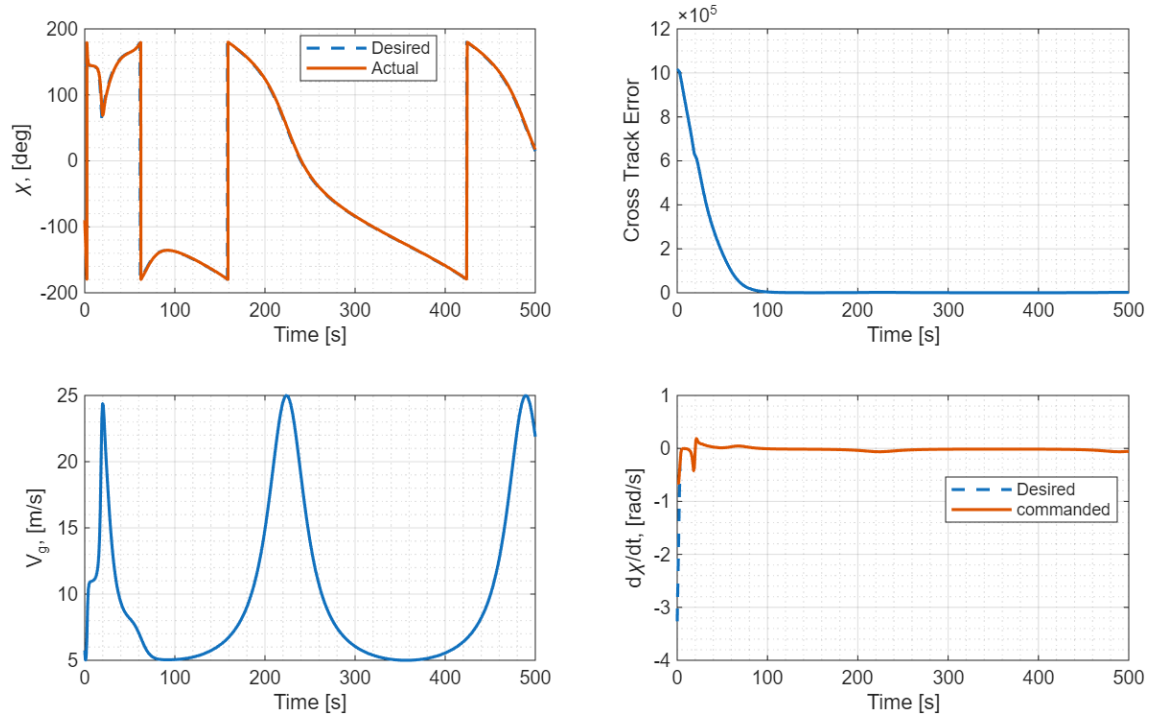
CASE I: Obstacle is completely outside The twice differentiable 2D obstacle in that case is given as,

$$C_2 : \quad g(x, y) = (x - 600)^2 + (y - 400)^2 - 100^2 \quad (2.24)$$

with initial position of the UAV as $P_0(900, -600) \text{ m}$ and all other parameters defined same as above in [2.6.1](#).



(a) Trajectory



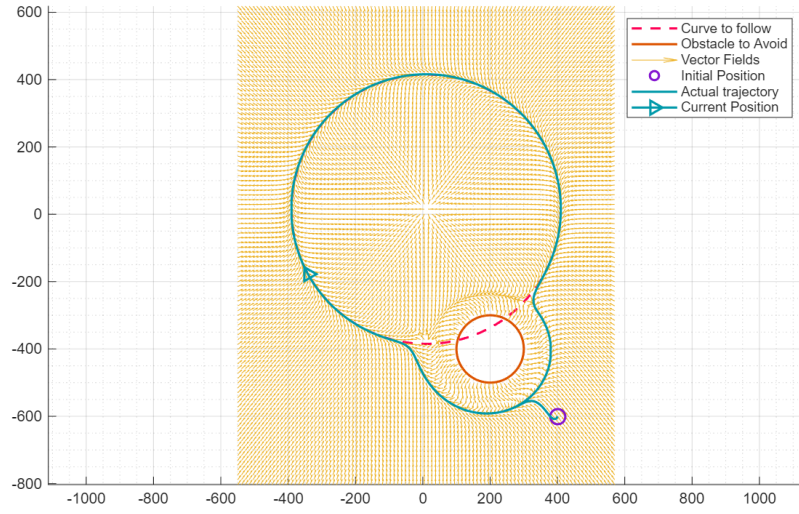
(b) Different parameters Variations with time for Singularity-free CVF guidance to a Circular desired path in presence of a circular obstacle

FIGURE 2.7: Path following and Obstacle avoidance with singularity-free CVF guidance for a circular desired path and a circular obstacle located completely outside the path

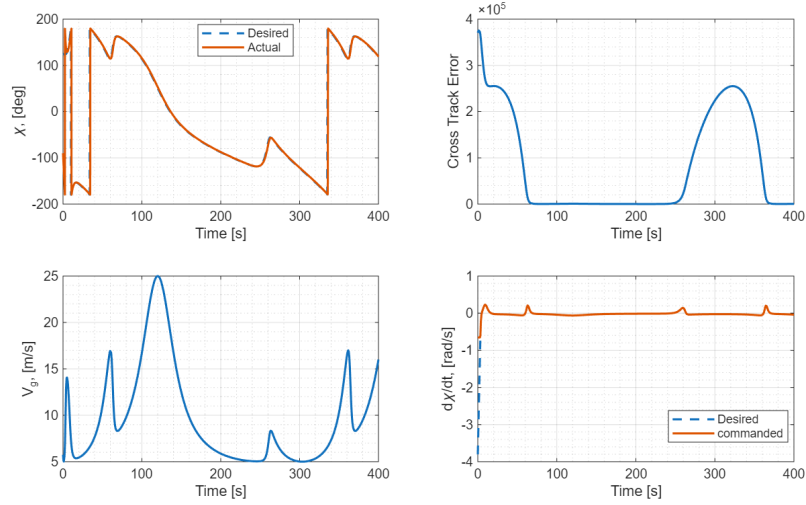
CASE II: Obstacle intersects the path The twice differentiable 2D obstacle in that case is given as,

$$C_2 : g(x, y) = (x - 400)^2 + (y + 600)^2 - 100^2 \quad (2.25)$$

with initial position of the UAV as $P_0(400, -600) m$ and all other parameters defined same as above in 2.6.1.



(a) Trajectory



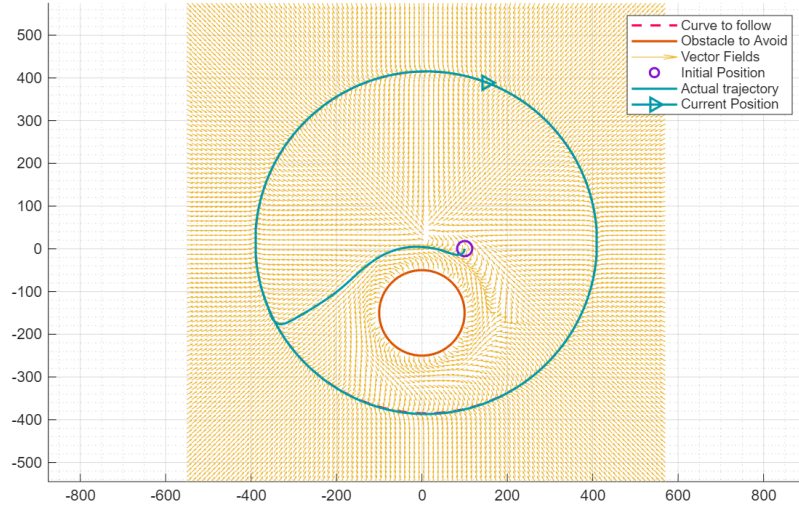
(b) Different parameters Variations with time for Singularity-free CVF guidance to a Circular desired path in presence of a circular obstacle

FIGURE 2.8: Path following and Obstacle avoidance with singularity-free CVF guidance for a circular desired path and a circular obstacle intersecting the desired path

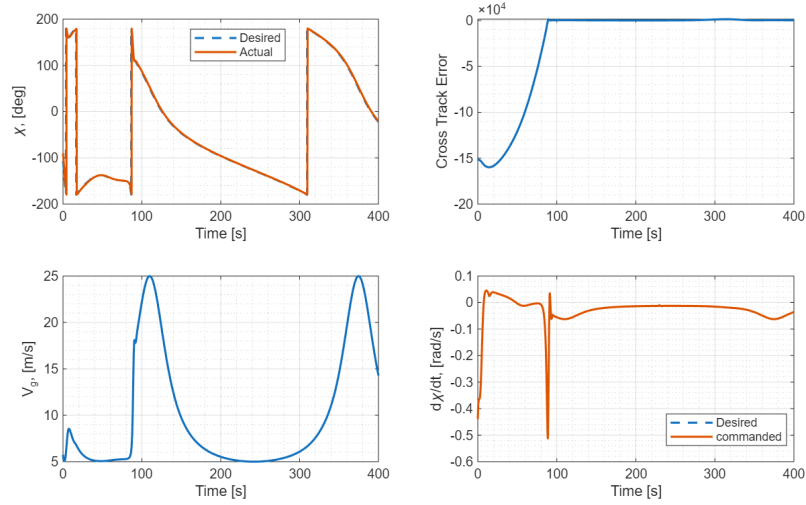
CASE III: Obstacle is inside The twice differentiable 2D obstacle in that case is given as,

$$C_2 : g(x, y) = x^2 + (y + 150)^2 - 100^2 \quad (2.26)$$

with initial position of the UAV as $P_0(100, 0) m$ and all other parameters defined same as above in 2.6.1.



(a) Trajectory



(b) Different parameters Variations with time for Singularity-free CVF guidance to a Circular desired path in presence of a circular obstacle

FIGURE 2.9: Path following and Obstacle avoidance with singularity-free CVF guidance for a circular desired path and a circular obstacle completely inside the desired path

2.6.2 Arbitrary Closed Desired Path and an Arbitrary Closed Obstacle

For the 2D path following of a twice differentiable arbitrary closed path with a twice differentiable arbitrary closed obstacle with singularity-free CVF-based guidance, the desired path is given by

$$\begin{aligned}
 C_1 : \quad f(x, y) = & \left(\frac{x}{k}\right)^2 + \left(\frac{y}{k}\right)^2 \\
 & + 0.6 \exp\left(-\left(\frac{x}{k} - 0.5\right)^2 - \left(\frac{y}{k} + 0.3\right)^2\right) \\
 & + 0.4 \exp\left(-\left(\frac{x}{k} + 0.7\right)^2 - \left(\frac{y}{k} - 0.6\right)^2\right) \\
 & + 0.35 \exp\left(-\left(\frac{x}{k} + 0.2\right)^2 - \left(\frac{y}{k} + 0.8\right)^2\right) \\
 & + 0.25 \exp\left(-\left(\frac{x}{k} - 0.9\right)^2 - \left(\frac{y}{k} - 0.1\right)^2\right) \\
 & + 0.15 \sin\left(\frac{3x}{k}\right) \cos\left(\frac{2y}{k}\right) + 0.12 \sin\left(\frac{5y}{k} + 0.7\right) \\
 & + 0.10 \cos\left(\frac{4x}{k} - 1.1\right) + \frac{0.08x}{k} \frac{y}{k} - 1.8
 \end{aligned}$$

where $k = 400$, with constant airspeed of 15 m/s , initial course angle $-\pi/2$, rotational sense of the solenoidal vector field for path-following ($s =$) is 1 (clock-wise sense), Path-following vector field design parameter ($\kappa =$) is 20, Obstacle vector field design parameter ($\kappa_{obs} =$) is 0.8, proportional feedback gain ($k_\chi =$) is 1.5 and the maximum allowable roll angle is 45° which limits the course rate. Additionally, simulation is done in the presence of wind $[6, 8] \text{ m/s}$.

CASE I: Obstacle is completely outside The twice differentiable 2D obstacle in that case is given as,

$$\begin{aligned}
C_2 : \quad g(x, y) = & \left(\frac{x - o(1)}{k} \right)^2 + \left(\frac{y - o(2)}{k} \right)^2 \\
& + 0.6 \exp \left(- \left(\frac{x - o(1)}{k} - 0.5 \right)^2 - \left(\frac{y - o(2)}{k} + 0.3 \right)^2 \right) \\
& + 0.4 \exp \left(- \left(\frac{x - o(1)}{k} + 0.7 \right)^2 - \left(\frac{y - o(2)}{k} - 0.6 \right)^2 \right) \\
& + 0.35 \exp \left(- \left(\frac{x - o(1)}{k} + 0.2 \right)^2 - \left(\frac{y - o(2)}{k} + 0.8 \right)^2 \right) \\
& + 0.25 \exp \left(- \left(\frac{x - o(1)}{k} - 0.9 \right)^2 - \left(\frac{y - o(2)}{k} - 0.1 \right)^2 \right) \\
& + 0.15 \sin \left(\frac{3(x - o(1))}{k} \right) \cos \left(\frac{2(y - o(2))}{k} \right) + 0.12 \sin \left(\frac{5(y - o(2))}{k} + 0.7 \right) \\
& + 0.10 \cos \left(\frac{4(x - o(1))}{k} - 1.1 \right) + \frac{0.08(x - o(1))}{k} \frac{(y - o(2))}{k} - 1.8
\end{aligned}$$

where $k = 100$ and $o = [700, -500]^T$, with initial position of the UAV as $P_0(1000, -700) m$ and all other parameters defined same as above in 2.6.2.

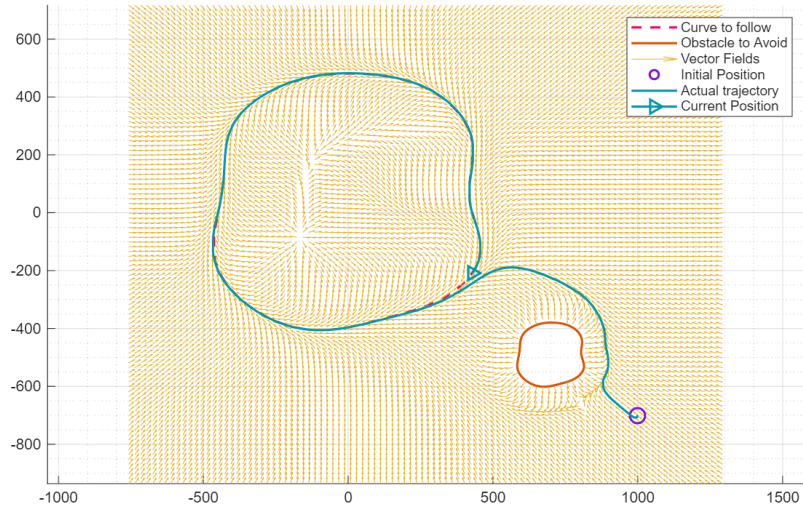


FIGURE 2.10: Trajectory for Path following and Obstacle avoidance with singularity-free CVF guidance for an arbitrary closed desired path and an arbitrary closed obstacle located completely outside the path (Part I)

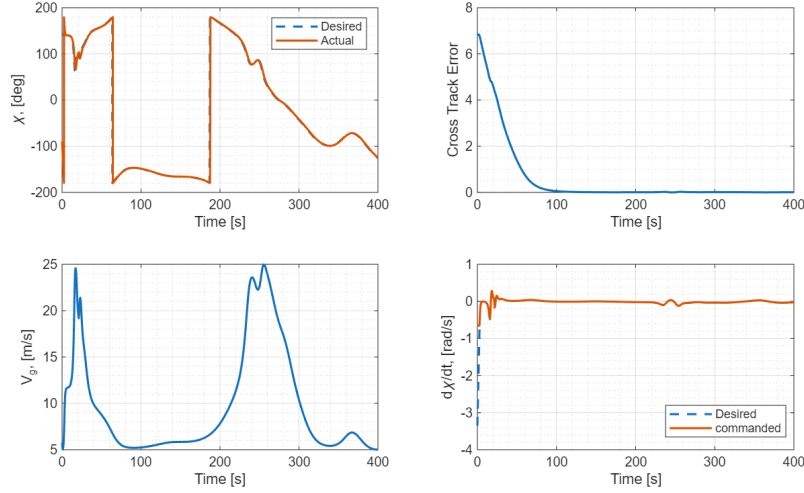


FIGURE 2.11: Different parameters Variations with time for Path following and Obstacle avoidance with singularity-free CVF guidance for an arbitrary closed desired path and an arbitrary closed obstacle located completely outside the path (Part II)

CASE II: Obstacle intersects the path The twice differentiable 2D obstacle in that case is given as,

$$\begin{aligned}
 C_2 : \quad g(x, y) = & \left(\frac{x - o(1)}{k} \right)^2 + \left(\frac{y - o(2)}{k} \right)^2 \\
 & + 0.6 \exp \left(- \left(\frac{x - o(1)}{k} - 0.5 \right)^2 - \left(\frac{y - o(2)}{k} + 0.3 \right)^2 \right) \\
 & + 0.4 \exp \left(- \left(\frac{x - o(1)}{k} + 0.7 \right)^2 - \left(\frac{y - o(2)}{k} - 0.6 \right)^2 \right) \\
 & + 0.35 \exp \left(- \left(\frac{x - o(1)}{k} + 0.2 \right)^2 - \left(\frac{y - o(2)}{k} + 0.8 \right)^2 \right) \\
 & + 0.25 \exp \left(- \left(\frac{x - o(1)}{k} - 0.9 \right)^2 - \left(\frac{y - o(2)}{k} - 0.1 \right)^2 \right) \\
 & + 0.15 \sin \left(\frac{3(x - o(1))}{k} \right) \cos \left(\frac{2(y - o(2))}{k} \right) + 0.12 \sin \left(\frac{5(y - o(2))}{k} + 0.7 \right) \\
 & + 0.10 \cos \left(\frac{4(x - o(1))}{k} - 1.1 \right) + \frac{0.08(x - o(1))}{k} \frac{(y - o(2))}{k} - 1.8
 \end{aligned}$$

where $k = 100$ and $o = [300, -300]^T$, with initial position of the UAV as $P_0(600, -400) m$ and all other parameters defined same as above in 2.6.2.

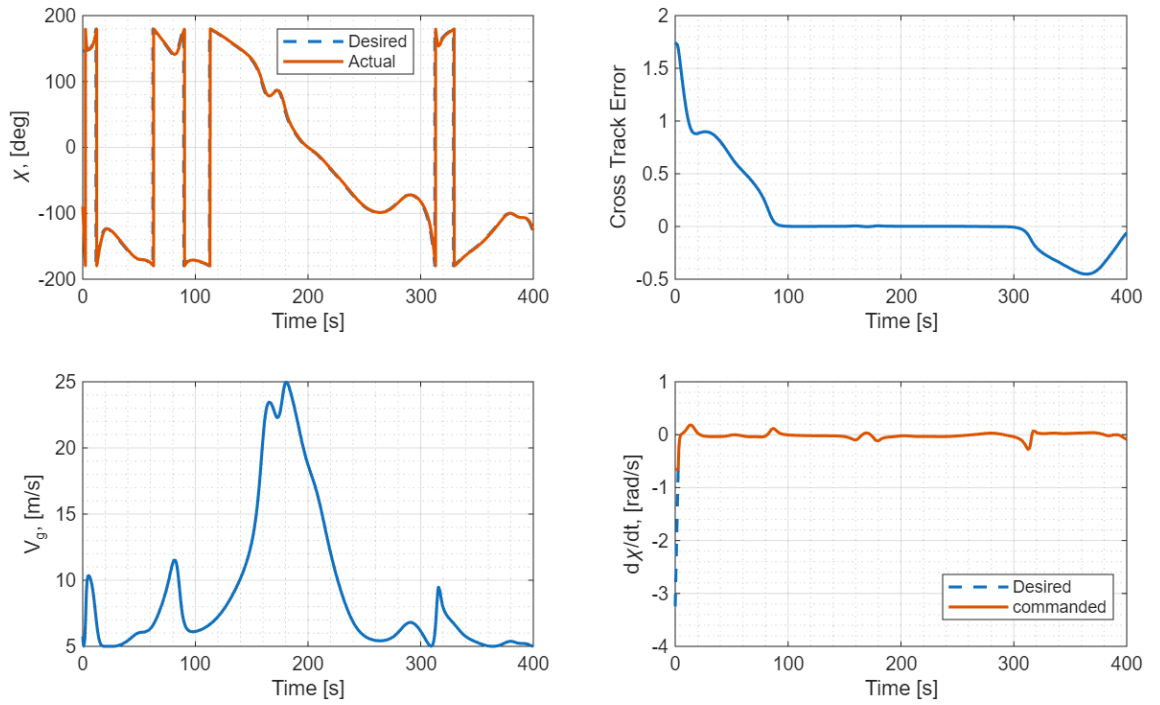
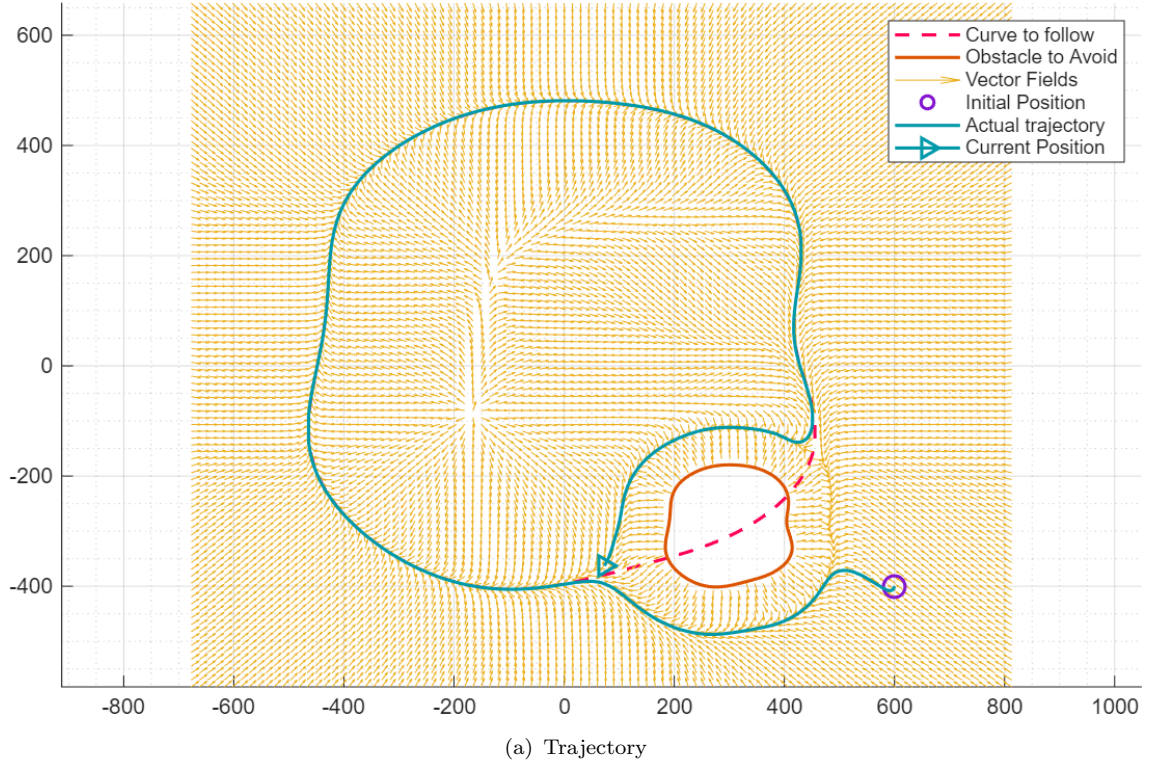
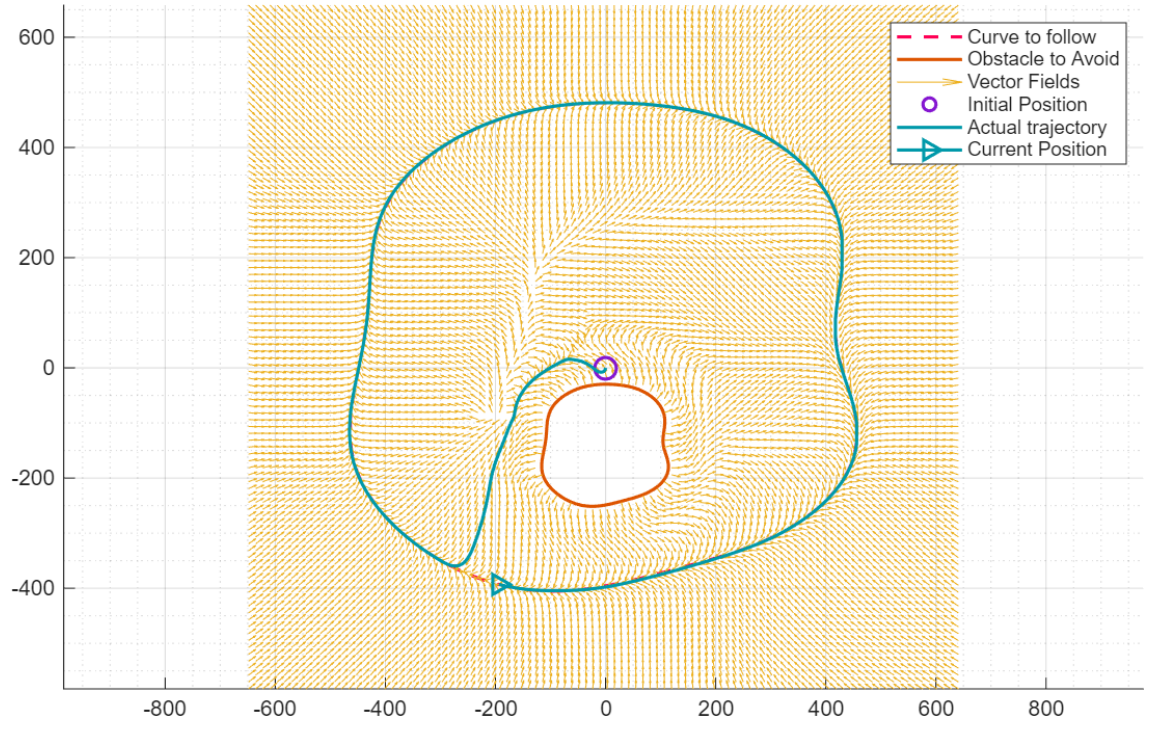


FIGURE 2.12: Path following and Obstacle avoidance with singularity-free CVF guidance for an arbitrary closed desired path and an arbitrary closed obstacle intersecting the desired path

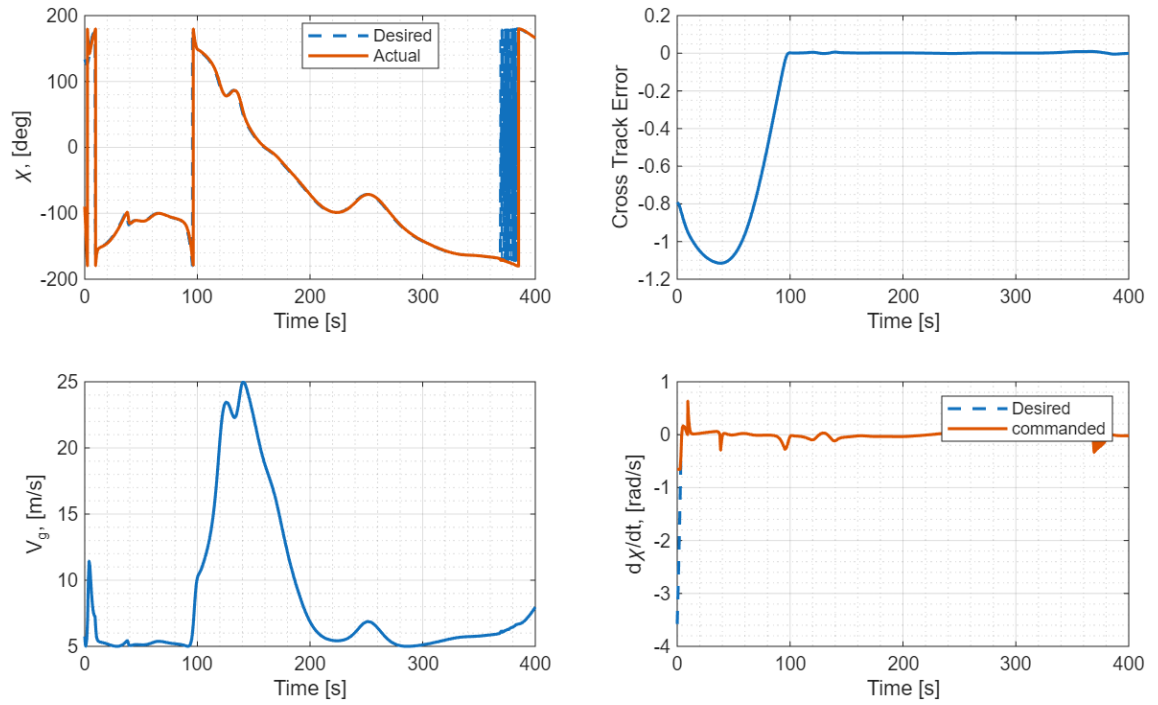
CASE III: Obstacle is inside The twice differentiable 2D obstacle in that case is given as,

$$\begin{aligned}
C_2 : \quad g(x, y) = & \left(\frac{x - o(1)}{k} \right)^2 + \left(\frac{y - o(2)}{k} \right)^2 \\
& + 0.6 \exp \left(- \left(\frac{x - o(1)}{k} - 0.5 \right)^2 - \left(\frac{y - o(2)}{k} + 0.3 \right)^2 \right) \\
& + 0.4 \exp \left(- \left(\frac{x - o(1)}{k} + 0.7 \right)^2 - \left(\frac{y - o(2)}{k} - 0.6 \right)^2 \right) \\
& + 0.35 \exp \left(- \left(\frac{x - o(1)}{k} + 0.2 \right)^2 - \left(\frac{y - o(2)}{k} + 0.8 \right)^2 \right) \\
& + 0.25 \exp \left(- \left(\frac{x - o(1)}{k} - 0.9 \right)^2 - \left(\frac{y - o(2)}{k} - 0.1 \right)^2 \right) \\
& + 0.15 \sin \left(\frac{3(x - o(1))}{k} \right) \cos \left(\frac{2(y - o(2))}{k} \right) + 0.12 \sin \left(\frac{5(y - o(2))}{k} + 0.7 \right) \\
& + 0.10 \cos \left(\frac{4(x - o(1))}{k} - 1.1 \right) + \frac{0.08(x - o(1))}{k} \frac{(y - o(2))}{k} - 1.8
\end{aligned}$$

where $k = 100$ and $o = [0, -150]^T$, with initial position of the UAV as $P_0(0, 0) m$ and all other parameters defined same as above in [2.6.2](#).



(a) Trajectory



(b) Different parameters Variations with time for Singularity-free CVF guidance to an arbitrary closed desired path in presence of an arbitrary closed obstacle

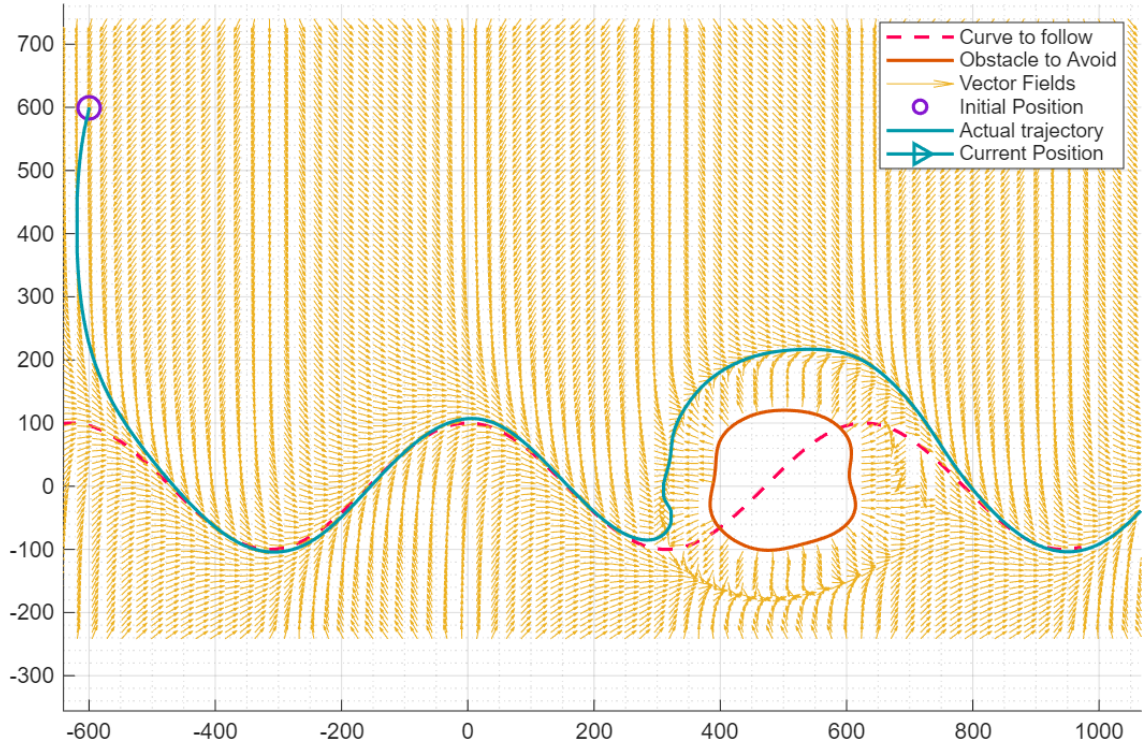
FIGURE 2.13: Path following and Obstacle avoidance with singularity-free CVF guidance for an arbitrary closed desired path and an arbitrary closed obstacle completely inside the desired path

2.6.3 Sine Path and An Arbitrary Closed Obstacle

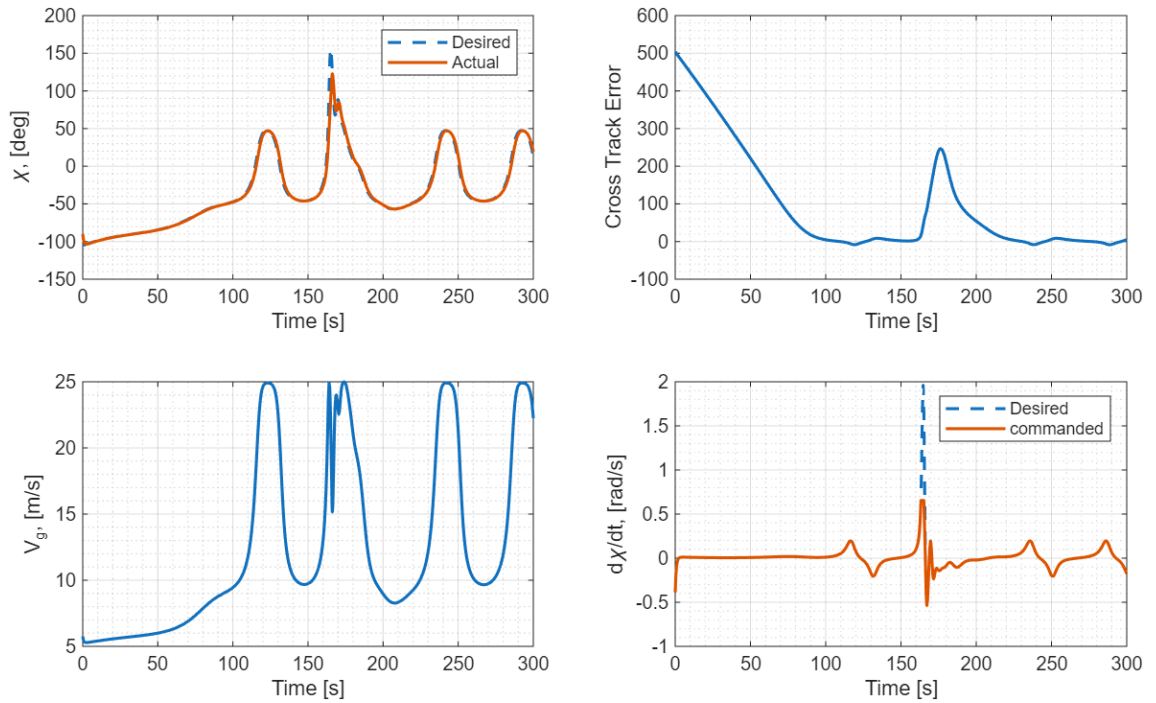
For the 2D path following of a twice differentiable sine path with a twice differentiable arbitrary closed obstacle with singularity-free CVF-based guidance, the desired path is given by

$$\begin{aligned}
 C_1 : \quad f(x, y) &= y - 100 \cos\left(\frac{x}{100}\right) \\
 C_2 : \quad g(x, y) &= \left(\frac{x - o(1)}{k}\right)^2 + \left(\frac{y - o(2)}{k}\right)^2 \\
 &+ 0.6 \exp\left(-\left(\frac{x - o(1)}{k} - 0.5\right)^2 - \left(\frac{y - o(2)}{k} + 0.3\right)^2\right) \\
 &+ 0.4 \exp\left(-\left(\frac{x - o(1)}{k} + 0.7\right)^2 - \left(\frac{y - o(2)}{k} - 0.6\right)^2\right) \\
 &+ 0.35 \exp\left(-\left(\frac{x - o(1)}{k} + 0.2\right)^2 - \left(\frac{y - o(2)}{k} + 0.8\right)^2\right) \\
 &+ 0.25 \exp\left(-\left(\frac{x - o(1)}{k} - 0.9\right)^2 - \left(\frac{y - o(2)}{k} - 0.1\right)^2\right) \\
 &+ 0.15 \sin\left(\frac{3(x - o(1))}{k}\right) \cos\left(\frac{2(y - o(2))}{k}\right) + 0.12 \sin\left(\frac{5(y - o(2))}{k} + 0.7\right) \\
 &+ 0.10 \cos\left(\frac{4(x - o(1))}{k} - 1.1\right) + \frac{0.08(x - o(1))}{k} \frac{(y - o(2))}{k} - 1.8
 \end{aligned}$$

where $k = 100$, and $o = [500, 0]^T$, with constant airspeed of 15 m/s , initial position of the UAV as $P_0(-600, 600) \text{ m}$, initial course angle $-\pi/2$, rotational sense of the solenoidal vector field for path-following ($s =$) is 1 (clock-wise sense), Path-following vector field design parameter ($\kappa =$) is 5, Obstacle vector field design parameter ($\kappa_{obs} =$) is 0.8, proportional feedback gain ($k_\chi =$) is 1.5 and the maximum allowable roll angle is 45° which limits the course rate. Additionally, simulation is done in the presence of wind $[6, 8] \text{ m/s}$.



(a) Trajectory



(b) Different parameters Variations with time for Path following and Obstacle avoidance with singularity-free CVF guidance for a sine desired path and an arbitrary closed obstacle

FIGURE 2.14: Different parameters Variations with time for Path following and Obstacle avoidance with singularity-free CVF guidance for a sine desired path and an arbitrary closed obstacle intersecting the desired path

Chapter 3

Conclusion

This thesis presented a Combined Vector Field (CVF)-based framework for the problem of path following and obstacle avoidance of an unmanned aerial vehicle in a two-dimensional environment. The desired path and obstacle boundaries were represented using twice differentiable implicit functions, enabling a unified and geometry-driven formulation of guidance laws.

A CVF-based path-following vector field was constructed by combining conservative (gradient-based) and circulatory (orthogonal) components using nonlinear weighting functions. This formulation ensures convergence toward the desired path while maintaining smooth motion along it. A similar CVF formulation was developed for obstacle avoidance, generating repulsive and circulatory behavior around obstacle boundaries.

To address the issue of singularities arising from direct vector field superposition, a structured blending approach based on guidance direction (course angle) combination was proposed. A distance-dependent weighting function was introduced to enable smooth transition between path-following and obstacle-avoidance behaviors. Additionally, a heuristic strategy for selecting the rotational direction of the obstacle

vector field and controlling angle wrapping was developed to reduce the likelihood of undesirable behaviors such as loops and premature stable points.

The proposed framework was implemented and evaluated through MATLAB simulations across a variety of scenarios, including circular paths, arbitrary closed curves, and sinusoidal paths in the presence of circular and non-circular obstacles. The results demonstrate that the approach is capable of achieving effective path following while avoiding obstacles under different geometric configurations and in the presence of wind disturbances. In most cases, the UAV successfully converges to the desired path after maneuvering around obstacles, indicating the practical viability of the proposed method.

However, several limitations of the approach were identified. The performance of the vector field is sensitive to design parameters such as the weighting factor and switching parameter, and appropriate tuning is required for different path and obstacle geometries. In certain edge cases, particularly when the obstacle intersects the desired path, artifacts in the vector field and undesired behaviors such as local loops or premature stable points may arise. Furthermore, the reliance on angle-based blending introduces sensitivity to angle wrapping and bifurcation behavior, which can affect robustness.

Future work can focus on addressing these limitations by developing a more systematic approach for parameter selection and by improving the robustness of the field combination strategy. In particular, methods that avoid reliance on angle wrapping or that incorporate constraint-based formulations may provide stronger guarantees on safety and convergence. Extension of the proposed framework to three-dimensional path following, as well as integration and simulation with UAV dynamics and experimental validation of this approach remains.

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